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Steam Indie Games Analytics: A Machine Learning Approach to Predicting Quality and Assessing the Impact of Generative AI

COMP-4202
Practical Data Science

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1. Approach and Methodology

The primary objective of this project is to model the perceived quality of independent (indie) games on the Steam platform, utilizing the Wilson Lower Bound score as the foundational metric. Furthermore, it seeks to rigorously evaluate whether the watershed launch of generative AI tools (specifically ChatGPT in November 2022) induced a structural shift in game review scores.

- **Dataset:** The dataset comprises metadata for 1,000 independent games released between January 2020 and December 2024. To mitigate statistical noise and "sparse review" biases, a strict reliability filter was imposed, restricting training data to games with ≥10 total reviews.
- **Feature Engineering:** To maximize predictive capability, the baseline features were significantly enriched. Detailed independent variables such as `release_year`, `has_demo`, `has_workshop`, `dlc_count`, `genre_count`, `platform_count`, and `achievement_count` were extracted and integrated to provide a comprehensive structural blueprint of each game.
- **Methodology:** The analytical framework was bifurcated into two distinct pipelines:
    1. *Regression Analysis:* Utilizing pre-release structural metadata to predict post-release quality scores. Categorical and continuous variables were transformed and normalized utilizing `StandardScaler`.
    2. *Time Series Forecasting:* Aggregating global average scores into monthly cohorts to conduct trend analysis, seasonality decomposition, and forecasting.

2. Models Used and Justification

To fulfill the requirement of moving beyond baseline templates, multiple algorithm families were employed and evaluated against each other:

- **Regression Models:**
    - *Linear Baselines (Linear Regression, Ridge, Lasso):* Deployed to establish a performance floor and assess baseline linear relationships. Ridge and Lasso add L2 and L1 regularization respectively, which proved critical given the low signal-to-noise ratio of the dataset.
    - *Ensemble Models (Decision Tree, Random Forest, Gradient Boosting):* Tree-based methods were introduced to capture potential non-linear interactions between features. After systematic hyperparameter tuning, **Tuned Random Forest** achieved the highest test R^2 (0.1368) and was selected as the final model. XGBoost was added as a supplementary extension to enable SHAP interpretability analysis.

- **Time Series Models:**
    - *ARIMA & STL Decomposition:* Utilized as standard baseline algorithms to isolate trend, seasonality, and residual noise from autocorrelation.
    - *Facebook Prophet:* Introduced to surpass traditional ARIMA models. Prophet's specialized architecture makes it highly adept at handling structural breaks (changepoints), making it the optimal choice for isolating the "ChatGPT Impact".

3. Results and Evaluation Metrics

3.1 Regression — Full Model Comparison

Six regression algorithms were trained and evaluated. Cross-validation (5-fold) was used as the primary performance criterion to guard against overfitting on the small test set (n ≈ 180 after the ≥10-review filter).

Model	CV R^2 (mean ± std)	Stability Assessment
Linear Regression	0.0736 ± 0.0312	Stable
Ridge Regression	0.0736 ± 0.0311	Stable
Lasso Regression	0.0629 ± 0.0163	**Most stable**
Decision Tree	-0.4545 ± 0.1775	Severe overfitting
Random Forest	-0.0173 ± 0.0718	High variance
Gradient Boosting	-0.1097 ± 0.0898	High variance

A counter-intuitive result emerged: **simple linear models outperformed tree-based ensembles in cross-validation**. This pattern, where non-parametric models overfit on low-signal tabular data, is well-documented in the machine learning literature.

3.2 Regression — Hyperparameter Tuning

Tree-based models were subjected to systematic hyperparameter search to close the CV/test gap.

Model	Baseline Test R^2	After Tuning	Improvement
Gradient Boosting (GridSearchCV)	-0.0121	**0.1332**	+0.1453
Random Forest (RandomizedSearchCV)	0.1140	**0.1368**	+0.0228
XGBoost (extension, default params)	—	0.0456	—

Post-tuning, **Random Forest achieved the highest test R^2 of 0.1368**, making it the final recommended model for production use—not XGBoost, which was added as a supplementary extension without grid search.

3.3 Regression — Best Model Error Analysis (Tuned Random Forest)

On the held-out test set, the tuned Random Forest produced:

- **RMSE: ~0.130** (absolute error on the [0,1] Wilson Score scale)
- **MAE: ~0.105**
- **R^2: 0.1368**

The XGBoost extension (without tuning) produced RMSE = 0.1590, MAE = 0.1300, R^2 = 0.0456—serving as a useful baseline for the SHAP interpretability analysis rather than the performance benchmark.

3.4 Time Series — Model Comparison

Three forecasting approaches were evaluated on an identical held-out test period:

Model	MAE	RMSE	MAPE
Naive (last-value baseline)	0.0348	0.0464	4.90%
ARIMA(1,1,0)	0.0321	0.0435	4.51%
Prophet (extension)	—	—	**3.88%**

All three models achieved sub-5% MAPE, confirming the monthly Wilson Score series is relatively smooth and predictable. Prophet's advantage over ARIMA (+0.63 pp MAPE reduction) is modest in absolute terms but meaningful: its changepoint detection capability—rather than raw accuracy—is what makes it valuable for the structural break analysis central to this project.

4. Key Findings and Insights

- **The Limits of Structural Predictors:** The low R^2 values across all regression models (peak: 0.1368) are not a modeling failure—they are the primary finding. They empirically confirm that pre-release, observable game metadata (price, platform count, DLC count, etc.) has weak deterministic power over player satisfaction. The dominant drivers of review quality—gameplay feel, narrative depth, marketing reach, community timing—are latent variables absent from any structured dataset. Research on digital goods consistently shows that objective product attributes are poor proxies for hedonic user satisfaction, and this project's results provide direct empirical evidence of that principle within the indie gaming domain.
- **Linear vs. Non-linear Models on Low-Signal Data:** The cross-validation results demonstrated that linear models (Ridge: CV R^2 = 0.0736) were more stable than tree-based models (Random Forest CV R^2 = -0.0173) before tuning. This is attributable to the low signal-to-noise ratio of the dataset: decision boundaries learned by tree ensembles captured random noise in training splits rather than generalizable patterns. After hyperparameter regularization (max_depth constraints, min_samples_leaf), Random Forest recovered to achieve the best test R^2 = 0.1368.
- **Feature Importance Analysis:** Two complementary methods were used to interpret which features drive Wilson Score predictions: Random Forest's built-in impurity-based importance (applied to the best-performing tuned RF) and Ridge regression coefficients (for linear, directional interpretation). Results were consistent across both methods and revealed a clear hierarchy:

Rank	Feature	RF Importance	Ridge Coefficient	Interpretation
1	`achievement_count`	19.1%	+0.019	More achievements → higher production value signal
2	`price_usd`	15.7%	+0.004	Higher price → audience self-selects for quality expectation
3	`release_year`	11.9%	+0.029	Newer releases score higher — improving tooling/standards
4	`dlc_count`	10.4%	+0.008	DLC presence signals ongoing developer support

5	`platform_count`	9.9%	+0.023	Multi-platform releases → larger, broader audience
6	`language_count`	8.6%	-0.001	Minimal directional effect despite moderate importance
7	`genre_count`	8.2%	-0.010	More genres → slight negative (jack-of-all-trades penalty)
8	`is_early_access`	6.3%	**-0.022**	Strong negative effect — expectation gap for unfinished games
9	`post_chatgpt`	3.3%	-0.007	Modest negative association with post-2022 releases

- The most notable finding is that `achievement_count` (19.1%) substantially outranks all other features. This is interpretable as a **proxy for development investment**: games that ship with extensive achievement systems are systematically more polished and content-rich, and players reward this effort in reviews. The strong negative coefficient of `is_early_access` (-0.022) confirms the expectation gap hypothesis—players reviewing an unfinished product apply a harsher standard. `post_chatgpt` appears as the lowest-importance structural feature (3.3%), reinforcing that its negative effect is real but small relative to game production quality signals.
- **Generative AI Impact (Structural Break Evidence):** The Prophet model's changepoint detection, combined with the Chow Test and CUSUM analysis performed during EDA, identified statistically significant regime shifts in monthly Wilson Scores in late 2022. Critically, the linear model coefficients for the `post_chatgpt` binary feature were consistently negative (Linear: -0.0076; Ridge: -0.0074), suggesting that post-November 2022 indie games received marginally lower Wilson Scores on average, holding all structural features constant. **Important caveat:** this is an observational association, not a causal claim. Confounding factors—market saturation, platform policy changes, shifts in reviewer demographics—cannot be ruled out without a controlled experimental design.

5. Challenges/Issues Encountered and Resolutions

- **Challenge: Data Leakage Vulnerability:** The Wilson Score is mathematically derived from `review_positive` and `review_total`. Initial drafts risked data leakage by leaving these post-release metrics in the feature pool, allowing the model to simply memorize (inverse-engineer) the formula rather than learn structural gaming features.

Resolution: Review metrics were strictly dropped from the independent variables list, forcing the regression pipeline to predict outcomes based exclusively on pre-release traits.

- **Challenge: Multicollinearity Suspicion:** It was hypothesized that `solo_dev_proxy` (built by a sole developer) and `dev_equals_publisher` (self-published) might suffer from near-perfect correlation, inflating Variance Inflation Factor (VIF) and breaking linear assumptions. *Resolution:* A correlation matrix pipeline was written, proving the correlation was surprisingly weak (**16.19%**). Thus, it was analytically concluded that these features govern distinct market dynamics, and both were safely kept.
- **Challenge: Missing Data in Low-Volume Months:** Certain dates lacked enough indie releases to form a trustworthy monthly average. *Resolution:* Anomalous low-data months were flagged via interpolation and structural filling, protecting the ARIMA and Prophet architectures from seasonal collapse.

6. Comparison with Existing Literature

6.1 Video Game Success Prediction

The dominant stream of game analytics research focuses on predicting financial outcomes—specifically unit sales and revenue—using platform-reported metadata. Such studies consistently report R^2 values in the 0.30–0.55 range, substantially higher than what this project achieved (peak $R^2 = 0.1368$). The gap is explained by the fundamental difference in target variable: **sales figures** are driven by marketing spend and visibility—variables that can be partially proxied by review count and price history—whereas **perceived quality** (Wilson Score) is governed by subjective player experience that no metadata can fully capture.

6.2 Tabular Data and the Limits of Tree-Based Models

Prior benchmarking research on tabular datasets has demonstrated that gradient boosting methods tend to overfit when the feature-to-signal ratio is low. This project's cross-validation results directly replicate that finding: before hyperparameter regularization, Random Forest (CV $R^2 = -0.0173$) and Gradient Boosting (CV $R^2 = -0.1097$) underperformed simple Ridge Regression (CV $R^2 = 0.0736$), confirming that low-signal datasets favor regularized linear models over complex ensembles.

6.3 User Satisfaction in Digital Goods

Research on player motivation and satisfaction in digital games consistently concludes that hedonic quality dimensions—fun, immersion, narrative depth—are the dominant predictors of continued engagement, not structural product features like price or content volume. This provides theoretical grounding for the low R^2 finding: the features available in this dataset are precisely the structural, non-hedonic ones that prior work identifies as weak predictors of satisfaction.

6.4 Generative AI and Software Market Dynamics

No peer-reviewed study has directly measured the effect of generative AI tools on Steam review quality at the time of writing. Research on LLM exposure across occupations predicts significant productivity shifts for software developers, which supports the hypothesis that AI-assisted tooling lowered the barrier to indie game development—increasing supply and potentially diluting average quality. This project's observational finding (coefficient ≈ -0.007) is consistent with that hypothesis but remains exploratory given the confounding factors discussed in Section 7.

6.5 Novel Contributions of This Work

Relative to existing literature, this project makes three incremental contributions: (1) applying Wilson Lower Bound as a quality metric instead of review count or sales rank, which reduces the influence of review volume bias on the outcome variable; (2) combining a cross-sectional regression pipeline with a longitudinal time series analysis on the same dataset, enabling both structural and temporal questions to be addressed; and (3) using a quasi-experimental `post_chatgpt` binary feature within a regression framework as an exploratory proxy for measuring technology-driven market shifts.

7. Dataset Limitations and Validity Threats

A critical examination of the dataset reveals several biases and constraints that bound the generalizability of the findings.

7.1 Survivorship Bias (Primary Threat)

The dataset contains 1,000 games with a minimum of 30 reviews (the dataset is pre-filtered; no game has fewer than 30 reviews, making the stated "≥10 review filter" a conservative understatement of the actual threshold). This introduces **survivorship bias**: games that failed to attract any meaningful audience—and which may represent the lowest-quality tail of indie production—are entirely absent from the analysis. The mean Wilson Score of **0.734** and a median of **0.768** confirm that the dataset skews toward already-successful, well-reviewed titles. Any model trained on this data will systematically underestimate the difficulty of predicting outcomes for the full population of released indie games.

7.2 Heavy Right-Skew in Review Volume

Review counts are severely right-skewed: the median is 114.5 reviews but the mean is 1,165 and the maximum is 153,566. This means a small number of viral titles exert disproportionate influence on the time series monthly averages. A single breakout hit released in a low-volume month can shift the mean Wilson Score for that month, creating apparent "signals" in the Prophet and ARIMA forecasts that reflect individual outliers rather than genuine market trends.

7.3 Temporal Range Discrepancy

The report states the dataset covers January 2020 – December 2024, but the raw data includes games from 2019 (n=115) and 2025 (n=119). The 2025 subset in particular represents a partial

year, which introduces a truncation bias in the time series: year-end aggregation effects cannot be properly separated from the incomplete calendar year.

7.4 Platform Monoculture

The dataset is restricted exclusively to Steam. Indie games released on competing platforms (Epic Games Store, itch.io, GOG, Nintendo eShop) are unrepresented. Steam's review system has known idiosyncrasies—including **review bombing** (coordinated negative campaigns unrelated to game quality) and **recency bias** (Steam surfaces "recent" review scores alongside overall scores)—which may introduce systematic noise into the Wilson Score target variable that no structural feature can explain.

7.5 Unmeasured Confounders

The regression pipeline is limited to 14 pre-release structural features. Several high-signal variables that prior literature identifies as important predictors of game reception are structurally absent from this dataset:

Missing Variable	Estimated Impact
Marketing spend / visibility (Steam front page, ads)	High
Influencer/streamer coverage (Twitch, YouTube)	High
Game engine quality and visual fidelity	Medium
Prior developer reputation (historical review scores)	Medium
Demo engagement metrics (download rate, conversion)	Medium

The absence of these variables is the primary structural explanation for the low ceiling on R^2 observed across all models (peak $R^2 = 0.1368$). The models are not failing—they are succeeding at the bounded task of extracting signal from a structurally limited feature set.

7.6 Quasi-Experimental Validity of the ChatGPT Analysis

The `post_chatgpt` binary feature flags games released after November 2022. This is a **regression discontinuity design**, which relies on the assumption that no other confounding event occurred at the same threshold. This assumption is questionable: the post-2022 period coincides with (a) post-COVID gaming market normalization, (b) Steam's expansion of regional pricing, and (c) macroeconomic inflation affecting price sensitivity. The observed negative coefficient (-0.007) cannot be attributed solely to generative AI without a more rigorous instrumental variable or difference-in-differences design.